

8th semester

ENVIRONMENTAL CHEMISTRY

PAST PAPERS

2019

SHORT QUESTIONS

Q1. Define soil and its composition.

Soil is a complex mixture of organic matter, minerals, gases, liquids, and countless microorganisms that form the top layer of the Earth's surface.

Its composition typically includes:

Mineral Particles like **Sand:** Larger, gritty particles. **Silt:** Intermediate-sized, smoother particles. **Clay:** Smallest, fine particles that can hold water and nutrients.

Organic Matter: Decomposed plant and animal material, including humus, which provides nutrients and improves soil structure.

Water: The liquid portion of soil that contains dissolved nutrients and is essential for plant growth.

Air: Pores in the soil that contains oxygen and carbon dioxide necessary for root respiration and microbial activity.

Microorganisms: Bacteria, fungi, and other microorganisms that decompose organic matter and contribute to nutrient cycling.

Q2. What is meant by eutrophication?

Eutrophication is a process in which a body of water, such as a lake or pond, becomes enriched with excessive nutrients, primarily nitrogen and phosphorus. This enrichment leads to an overgrowth of algae and aquatic plants, resulting in decreased oxygen levels as these organisms decompose. The reduced oxygen can harm aquatic life, leading to fish kills and disruptions in the ecosystem. Eutrophication is often caused by human activities like agricultural runoff and sewage discharge. Monitoring and managing nutrient inputs are essential to prevent or mitigate eutrophication in aquatic ecosystems.

Q3. Differentiate between macronutrients and micronutrients.

Macronutrients and micronutrients are two categories of essential nutrients required by organisms, including plants and humans.

1. Macronutrients

Definition: Macronutrients are nutrients that are required in relatively large quantities by living organisms.

Types: The main macronutrients for plants and humans include carbohydrates, proteins, and fats. In the context of plants, macronutrients also include elements like nitrogen, phosphorus, and potassium.

Function: Macronutrients provide the energy necessary for growth, development, and maintenance of the body or plant structure.

2. Micronutrients

Definition: Micronutrients are nutrients that are needed in smaller amounts compared to macronutrients.

Types: Micronutrients include vitamins and minerals such as iron, zinc, copper, vitamins A, B, C, and others.

Function: While required in smaller quantities, micronutrients play crucial roles in various physiological and biochemical processes. They are essential for functions like enzyme activity, immune system support, and overall health.

Q4. How is environment affected by PCBs?

Polychlorinated Biphenyls have detrimental effects on the environment:

- PCBs are highly stable and resistant to degradation, leading to their long-term presence in the environment.
- PCBs can accumulate in aquatic and terrestrial organisms, particularly in fatty tissues, which leads to increased concentrations up the food chain.
- PCBs are toxic to a wide range of organisms, including fish, birds, and mammals, leading to impaired health and reduced reproductive success.
- PCBs can contaminate water bodies and soil, affecting ecosystems and potentially

Q5. How does the pH of soil affect the availability of nutrients?

The pH of soil plays a crucial role in nutrient availability. Soils with an **acidic pH** (below 7) can reduce the availability of essential nutrients such as phosphorus, calcium, and magnesium. Conversely, **alkaline soils** (pH above 7) can limit the accessibility of micronutrients like iron, manganese, and zinc. **Neutral soils** (pH around 7) generally promote optimal nutrient availability for most plants. Soil pH affects nutrient solubility, which in turn influences how effectively plants can uptake these nutrients. Adjusting soil pH through amendments like lime or sulfur can help optimize nutrient availability for specific crops.

Q6. Give at least four volatile organic compounds that pollute the soil.

Volatile Organic Compounds (VOCs) that can pollute soil include:

Benzene: A common pollutant from gasoline and oil spills, it can leach into the soil and contaminate groundwater.

Toluene: Toluene can infiltrate the soil through spills and leaks, leading to soil contamination.

Xylene: Xylene can easily penetrate soil from sources like industrial discharge, making it a soil pollutant.

Trichloroethylene (TCE): TCE is a chlorinated VOC used in various industrial processes. It is notorious for contaminating soil and groundwater due to improper disposal or leaks

Q7. How is biodiversity affected by the presence of pollutants?

The presence of pollutants can significantly impact biodiversity in various ways.

Pollutants can directly harm or kill organisms, leading to a decrease in species diversity. They can also disrupt food chains and ecosystems by affecting species interactions. Moreover, pollutants can lead to genetic mutations and reduced reproductive success, further diminishing biodiversity. Habitat degradation resulting from pollution can reduce available niches for species, causing shifts in biodiversity.

Q8. Define the term “ion exchange capacity of the soil”.

The Ion Exchange Capacity of soil refers to its ability to retain and exchange ions, specifically cations. It is a critical property that influences the soil's fertility and its capacity to hold essential nutrients for plants. CEC is determined by the presence of clay minerals and organic matter in the soil, which have negatively charged

surfaces capable of attracting and holding cations. Soils with a higher CEC can retain more nutrients, making them more fertile.

Q9. How is environmental monitoring different from environmental analysis?

Environmental monitoring: It involves continuous or periodic data collection from various environmental parameters, such as air quality, water quality, and biodiversity. It aims to track changes in the environment over time, detect trends, and provide early warnings of environmental issues.

Environmental analysis: It involves conducting scientific experiments, testing, and analysis of environmental samples, such as soil, water, or air, to assess pollution levels, toxicity, or the presence of contaminants.

Q10. How are micronutrients beneficial for plant growth?

Micronutrients, which include elements like **iron, manganese, zinc, copper, and boron**, are essential for plant growth and development. These elements serve as cofactors for various enzymes involved in critical metabolic processes within plants. They play a crucial role in photosynthesis, respiration, and nitrogen metabolism. Micronutrients are also involved in the formation of chlorophyll, the green pigment responsible for photosynthesis, and the production of proteins and nucleic acids, which are vital for plant growth.

LONG QUESTIONS

Q1. How are littering and mining helpful in increasing soil quality?

Littering and Soil Quality:

Negative Impact of Littering:

- Littering refers to the indiscriminate disposal of waste materials, often including non-biodegradable items like plastics, glass, and metals. These materials do not decompose easily, and some can release harmful substances when exposed to environmental conditions. When non-biodegradable litter accumulates in the environment, it can lead to soil pollution. Plastics, for example, can break down into microplastics, which are difficult to remove and can persist in the soil for extended periods. Chemicals from certain litter items can leach into the soil, affecting its quality.

Positive Impact of Organic Litter:

- On the other hand, organic litter, such as fallen leaves, dead plants, and natural debris, can be beneficial for soil quality. When these organic materials decompose, they add organic matter to the soil, which improves its structure, moisture retention, and nutrient content. This enhances the soil's fertility and supports plant growth.

Mining and Soil Quality:

Negative Impact of Mining:

- Mining activities involve the extraction of valuable minerals and ores from the Earth. This often necessitates the removal of large quantities of soil, particularly the topsoil layer, which is rich in organic matter and nutrients. The removal of topsoil can lead to soil erosion and degradation. Without this fertile layer, the soil becomes less suitable for plant growth, impacting its quality.

Contamination and Acid Mine Drainage:

- Mining can introduce pollutants and chemicals into the soil. For example, certain minerals and ores contain heavy metals like lead, mercury, and arsenic, which can leach into the soil, making it toxic to plants and organisms. Acid mine drainage is a severe issue associated with mining. It occurs when sulfide minerals are exposed to air and water during mining operations. This process generates acidic runoff that not only contaminates the soil but also nearby water bodies.

Q2. Explain soil erosion and some methods of its prevention in detail.

- Soil erosion is the process by which the topsoil of land is removed or displaced by various natural or human-induced factors, such as wind, water, or human activities. It is a significant environmental issue as it can lead to the loss of fertile topsoil, reduced agricultural productivity, and increased sedimentation in water bodies.

Causes of Soil Erosion:

a. Water Erosion: This is caused by rainfall and surface runoff. It includes sheet erosion (thin layers of soil are removed), rill erosion (small channels form), and gully erosion (larger, deeper channels).

b. Wind Erosion: Wind can carry away loose, dry soil particles, particularly in arid or semi-arid regions.

c. Tillage and Agriculture: Traditional ploughing and agricultural practices can leave soil exposed, making it vulnerable to erosion.

d. Deforestation: Removing trees and vegetation can destabilize soil and increase erosion risk.

e. Construction and Urbanization: Urban development can lead to increased runoff and erosion due to impermeable surfaces like roads and buildings.

Methods of Soil Erosion Prevention:

Terracing:

Terracing is a technique used in hilly or sloping areas. It involves building a series of flat, level steps, often with stone or earth walls, into the slope. These steps slow down the flow of water, preventing it from gaining enough energy to erode the soil. Terraces also provide level areas for planting crops. This method is common in regions with steep terrain.

Crop Rotation and Cover Crops:

Crop rotation involves changing the type of crops grown in a particular field over time. This practice helps maintain soil fertility, as different crops have different nutrient requirements. Cover crops, such as legumes and grasses, are planted between cash crops. They protect the soil during fallow periods and between planting seasons by preventing erosion, increasing soil organic matter, and reducing weed growth.

Reforestation:

Reforestation involves planting trees and vegetation in areas where natural forests have been cleared or degraded. Trees have deep root systems that anchor soil, reducing the risk of both water and wind erosion. They also enhance biodiversity and improve the overall environmental health of an area.

Contour Farming:

Contour farming is a practice where crops are planted along the contour lines of sloping terrain. By doing this, water runoff is slowed and prevented from eroding the soil. It helps to reduce the formation of gullies and rills on sloping lands.

Grassed Waterways:

Grassed waterways are designed channels with grass or other vegetation. They're used to manage and control runoff from fields, reducing soil erosion and sedimentation. These channels provide a path for water to flow without causing erosion.

Sediment Basins and Silt Fences:

In construction sites, sediment basins are temporary structures designed to trap sediment-laden runoff water. Silt fences are fabric barriers installed along construction perimeters. Both methods help contain sediment and prevent it from entering water bodies.

Soil Stabilizers:

Soil stabilizers include natural or synthetic materials applied to the soil surface to prevent erosion. They form a protective layer that shields the soil from wind and water. Mulching and using geotextiles are common practices.

Education and Regulation:

Promoting awareness about soil erosion and enforcing regulations related to land use and development practices is crucial. This involves educating landowners, farmers, and developers about the importance of erosion control and ensuring compliance with laws and guidelines designed to prevent erosion.

Sustainable Land Management:

Sustainable land management practices, such as agroforestry and organic farming, prioritize soil conservation. Agroforestry combines trees with crops or livestock to create more resilient and erosion-resistant ecosystems. Organic farming techniques minimize soil disturbance and promote the use of natural soil amendments to maintain soil health.

Q3. Explain the hazardous effects of aflatoxins on the environment. Especially on human health in detail.

- Aflatoxins are a group of naturally occurring mycotoxins produced by certain fungi, primarily *Aspergillus flavus* and *Aspergillus parasiticus*. These toxins can have hazardous effects on both the environment and human health. Here's a detailed explanation of their impact:

Environmental Effects:

a. Contamination of Crops: Aflatoxins are known to contaminate a wide range of crops, including peanuts, corn, cottonseed, and tree nuts. This contamination occurs in the field before harvest and can spread throughout the environment.

b. Soil and Water Pollution: Aflatoxin-producing moulds can survive in the soil, further contaminating the environment. Additionally, aflatoxins can leach into groundwater, affecting drinking water sources.

c. Toxicity to Wildlife: Aflatoxins can harm wildlife that consumes contaminated crops, affecting ecosystems. For example, birds and rodents that consume contaminated grains can suffer from aflatoxicosis, which can be fatal.

d. Agricultural and Economic Impact: Aflatoxin contamination can lead to significant economic losses for farmers and the agricultural industry due to reduced crop yields and rejected or downgraded produce.

Hazardous Effects on Human Health:

a. Carcinogenicity: Aflatoxins, especially Aflatoxin B1, are potent carcinogens. Prolonged exposure to these toxins through contaminated food can increase the risk of liver cancer. It's classified as a Group 1 carcinogen by the International Agency for Research on Cancer (IARC).

b. Liver Damage: Aflatoxins are primarily metabolized in the liver, where they can cause extensive damage. Chronic exposure can lead to liver cirrhosis, fibrosis, and other liver diseases.

c. Immunosuppression: Aflatoxins can weaken the immune system, making individuals more susceptible to various infections and diseases.

d. Growth Impairment: Children exposed to aflatoxins may experience growth and developmental issues. This can lead to stunting and malnutrition, especially in regions where aflatoxin contamination is prevalent.

e. Acute Aflatoxicosis: In cases of acute exposure to high levels of aflatoxins, individuals can experience severe symptoms like vomiting, abdominal pain, and jaundice. This can be fatal in extreme cases.

f. Developmental Effects: Aflatoxins may also pose risks to fetal development, potentially leading to birth defects or developmental problems.

Q4. Explain the concept of Green chemistry giving its principles.

- Green chemistry, also known as sustainable chemistry or environmentally friendly chemistry, is an approach to the design, development, and

application of chemical processes and products that aim to minimize their negative impact on the environment and human health. The concept of green chemistry revolves around finding innovative and efficient ways to carry out chemical reactions while reducing or eliminating the use and generation of hazardous substances.

The principles of green chemistry, as outlined by Paul Anastas and John Warner in 12 Principles of Green Chemistry, are as follows:

Prevention: It is better to prevent waste and pollution at the source rather than cleaning it up after it has been created. This principle encourages the design of chemical processes and products with reduced environmental impacts.

Atom Economy: This principle emphasizes maximizing the incorporation of all reactant atoms into the final product. Reactions with a high atom economy reduce waste generation.

Less Hazardous Synthesis: Green chemistry promotes the use of methods that are less hazardous, such as catalytic processes, as opposed to using toxic reagents.

Designing Safer Chemicals: Chemical products should be designed to be effective but with minimal toxicity and environmental impact. The aim is to reduce the negative consequences associated with their use.

Safer Solvents and Auxiliaries: Green chemistry encourages the use of environmentally benign solvents and auxiliary substances to minimize the environmental impact of chemical processes.

Design for Energy Efficiency: Chemical processes should be designed to be energy-efficient, reducing energy consumption and its associated environmental impact.

Use of Renewable Feedstocks: Whenever possible, green chemistry promotes the use of raw materials derived from renewable resources rather than fossil fuels or other finite sources.

Reduce Derivatives: Minimize the use of derivatization (temporary chemical modifications) whenever possible to decrease waste generation and simplify the synthesis process.

Catalysis: Employ catalytic processes whenever possible to increase efficiency and reduce waste generation. Catalysts enable the reaction to occur with lower temperatures and fewer by-products.

Design for Degradation: Chemical products should be designed in a way that they break down into innocuous substances after use to minimize environmental persistence.

Real-time Analysis for Pollution Prevention: Green chemistry promotes the use of real-time monitoring and control during chemical processes to prevent pollution and ensure efficient reactions.

Inherently Safer Chemistry for Accident Prevention: Design chemical processes to avoid accidents, explosions, and releases of hazardous materials. Safety considerations are a crucial aspect of green chemistry.

Q5. Explain the principal and Application of UV and visible spectroscopy for environmental Analysis.

- UV-visible spectroscopy is widely used in environmental analysis. It is based on the absorption of ultraviolet (UV) and visible light by molecules, which allows for the qualitative and quantitative analysis of various compounds.

Principle of UV-Visible Spectroscopy:

UV-visible spectroscopy operates on the principle that molecules absorb specific wavelengths of light in the UV and visible regions of the electromagnetic spectrum. When a sample is exposed to UV or visible light, certain electrons within the molecules can transition from their ground state to an excited state, absorbing energy in the process. The energy absorbed corresponds to specific wavelengths of light, and this absorption is detected by a spectrophotometer. The resulting spectrum is known as an absorption spectrum, which can provide valuable information about the composition and concentration of the analyte in the sample.

Applications of UV-Visible Spectroscopy for Environmental Analysis:

Water Quality Analysis: UV-visible spectroscopy is extensively used for water quality assessment. It helps determine the concentration of pollutants such as heavy metals, organic compounds, and disinfection by-products (DBPs). For example, the presence of nitrate, nitrite, and phosphates in water can be measured using UV-visible absorption.

Air Pollution Monitoring: UV-visible spectroscopy aids in the detection and quantification of air pollutants, such as ozone (O₃), nitrogen dioxide (NO₂), and volatile organic compounds (VOCs). These compounds can have adverse effects on human health and the environment, so monitoring their levels is crucial.

Analysis of Soil Contaminants: UV-visible spectroscopy can be employed to determine the presence of pollutants in soil samples, including heavy metals and organic contaminants. This helps in assessing soil quality and potential environmental risks.

Monitoring Environmental Reactions: UV-visible spectroscopy is utilized to study the kinetics of environmental reactions, such as the photodegradation of

pollutants in water and the degradation of atmospheric pollutants in the presence of sunlight.

Monitoring UV-Induced Damage: UV radiation can cause damage to the environment, and UV-visible spectroscopy can help in studying the effects of UV exposure on different materials and organisms.

Analysis of Natural Organic Matter: UV-visible spectroscopy is useful in characterizing natural organic matter in water bodies, such as dissolved organic carbon (DOC). This information aids in understanding the quality of natural water sources and their potential reactivity with other pollutants.

Monitoring Environmental Analytes in Real-Time: In recent years, portable UV-visible spectrophotometers have been developed, enabling real-time monitoring of environmental analytes in the field. This technology is particularly valuable for assessing water quality in remote or challenging locations.

Overall, UV-visible spectroscopy is a versatile and powerful technique for environmental analysis due to its simplicity, sensitivity, and ability to analyze a wide range of analytes. It plays a crucial role in understanding and mitigating environmental issues by providing valuable data for decision-making and policy implementation.

Q6. What is Atomic Absorption Spectroscopy and its role in environmental monitoring?

- Atomic Absorption Spectroscopy (AAS) is a powerful analytical technique used in chemistry to determine the concentration of specific elements in a sample. It is particularly valuable for measuring the concentration of metals and metalloids in various substances.

Working of Atomic Absorption Spectroscopy

Atomization: The sample is first converted into atomic vapour. This is typically done by heating the sample to a high temperature, causing the atoms to break apart from their chemical bonds and exist in a gaseous state.

Absorption: A light beam, usually from a hollow cathode lamp that emits light at the specific wavelength corresponding to the element of interest, is passed through the atomic vapour. The atoms in the vapour will absorb light at characteristic wavelengths specific to the element.

Detection: The light that passes through the vapour is then measured by a detector. The amount of light absorbed is directly proportional to the concentration of the element present in the sample.

The concentration of the element is determined by comparing the amount of light absorbed to a calibration curve, which is created using known standards of the element at various concentrations.

Role in Environmental Monitoring:

Atomic Absorption Spectroscopy plays a crucial role in environmental monitoring for several reasons:

Detecting Trace Metals: AAS is highly sensitive and capable of detecting trace amounts of metals in complex environmental samples. Many metals, such as lead, mercury, arsenic, cadmium, and chromium, can be harmful to human health and the environment even at very low concentrations. AAS enables the detection and quantification of these metals to assess environmental contamination and potential risks.

Monitoring Water Quality: AAS is extensively used to analyze water samples from various sources, including rivers, lakes, groundwater, and wastewater. It helps determine the presence and concentration of metals in water, which is essential for assessing water quality and potential pollution sources.

Soil and Sediment Analysis: In environmental studies, AAS is employed to analyze soil and sediment samples. It aids in identifying the levels of metals present in the soil, which can have implications for agricultural practices, land use, and potential contamination issues.

Air Pollution Monitoring: AAS is utilized in air quality monitoring to measure the levels of metallic pollutants in the atmosphere. This information is essential for assessing the impact of industrial activities, traffic emissions, and other sources of pollution on air quality and human health.

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2020

SHORT QUESTIONS

Q1. Toxicity of Cr is attribute to Cr(IV) not to Cr(III). Why?

The toxicity of chromium is indeed attributed to the hexavalent form, Cr(IV), rather than the trivalent form, Cr(III). Here's why:

Chemical Reactivity: Cr(III) is generally less reactive than Cr(IV). Cr(IV)

compounds are more likely to form highly reactive free radicals that can damage biological molecules like DNA, proteins, and lipids.

Cellular Uptake: Cr(IV) is more easily taken up by cells than Cr(III). Once inside cells, Cr(IV) can undergo redox reactions, leading to the generation of reactive oxygen species (ROS), which contribute to oxidative stress and cellular damage.

Bioavailability: Cr(IV) is more soluble and bioavailable than Cr(III), making it easier for the former to enter biological systems and exert its toxic effects.

Q2. What is CEC why does soil hold only cations?

Cation Exchange Capacity(CEC) is a fundamental property of soil that measures its ability to hold and exchange cations (positively charged ions). Soil's CEC is determined by the presence of negatively charged sites on clay minerals and organic matter. These negatively charged sites attract and retain cations, making them available to plants for nutrient uptake. The soil can hold cations because of the electrostatic attraction between the negative charges on the soil particles and the positive charges on the cations. While soil primarily holds cations due to its negative charge, it can also retain some anions (negatively charged ions) through processes like anion exchange, but the majority of nutrient uptake by plants relies on the attraction and exchange of cations.

Q3. What are the major causes of soil erosion?

Major causes of soil erosion include:

Water Erosion: This occurs due to rainfall, runoff, and the flow of rivers and streams, leading to the removal of topsoil.

Wind Erosion: Strong winds can carry away loose, dry soil particles, particularly in arid and semi-arid regions, causing desertification.

Human Activities: Deforestation, agriculture without proper soil conservation practices, construction, and mining can accelerate soil erosion.

Soil Compaction: Soil compaction from heavy machinery or overgrazing reduces soil structure, making it more susceptible to erosion.

Natural Factors: Earthquakes, volcanic eruptions, and landslides can expose soil to erosion, especially on steep slopes.

Q4. What are aflatoxins?

Aflatoxins are highly toxic and carcinogenic secondary metabolites produced by certain moulds, primarily *Aspergillus* species, that contaminate crops like peanuts, corn, and tree nuts. These mycotoxins pose serious health risks when ingested or inhaled. They can contaminate food products at various stages, from cultivation to storage, and are especially prevalent in warm and humid climates.

Q5. Discuss control measures for water erosion.

To control water erosion, various measures can be implemented to reduce the impact of rainfall and flowing water on the soil. Here are some effective control measures:

Vegetative Cover: Planting cover crops, grass, or trees helps to create a protective layer that shields the soil from the impact of raindrops and slows down water runoff. The roots also bind the soil together, reducing erosion.

Cover Cropping: Planting cover crops during periods when the main crop is not growing provides continuous cover for the soil. Cover crops protect against erosion, enhance soil structure, and add organic matter.

Erosion Control Blankets: Using erosion control blankets or mats made of natural or synthetic materials helps stabilize the soil surface and protect it from the impact of rainfall.

Q6. Differentiate between macro and micronutrients in soil.

Q7. Describe the principle of UV-visible spectroscopy.

UV-visible spectroscopy operates on the principle that molecules absorb specific wavelengths of light in the UV and visible regions of the electromagnetic spectrum. When a sample is exposed to UV or visible light, certain electrons within the molecules can transition from their ground state to an excited state, absorbing energy in the process. The energy absorbed corresponds to specific wavelengths of light, and this absorption is detected by a spectrophotometer. The resulting spectrum is known as an absorption spectrum, which can provide valuable information about the composition and concentration of the analyte in the sample.

Q8. Discuss the main sources of lead in the environment.

The main sources of lead in the environment include:

Lead-Based Paint: Lead-based paint, used extensively in older homes, can deteriorate over time, leading to lead dust and chips that can contaminate soil and indoor environments.

Industrial Activities: Industries like mining, smelting, and battery manufacturing release lead emissions into the air and water, contributing to environmental contamination.

Vehicle Emissions: While leaded gasoline has been phased out in many countries, past use of leaded gasoline has left residues in soil and dust along roadways.

Lead Pipes: Water systems with lead pipes or lead solder can leach lead into drinking water, posing health risks, especially in older infrastructure.

Consumer Products: Various consumer goods, like imported toys, jewellery, and ceramics, may contain lead-based components that can release lead if ingested or in contact with humans.

Q9. Why PCBs were banned?

Polychlorinated Biphenyls (PCBs) were banned due to their severe environmental and health risks. PCBs are toxic, persistent organic pollutants that do not readily break down in the environment. They accumulate in the food chain, leading to bioaccumulation in animals and, ultimately, in humans. PCBs have been linked to a range of health problems, including cancer, reproductive issues, and developmental abnormalities.

Q10. What is atom economy?

Atom economy is a concept in chemistry *that measures the efficiency of a chemical reaction in utilizing reactant atoms to form desired products*. It quantifies the percentage of atoms from the reactants that end up in the desired product, thus minimizing waste and maximizing resource utilization. A higher atom economy signifies a more sustainable and efficient reaction. It is calculated by dividing the total molecular weight of the desired product by the sum of the molecular weights of all the reactants. Atom economy is an important principle in green chemistry, promoting the design of processes that minimize waste generation and environmental impact.

LONG QUESTIONS

Q1. What is soil pollution? What are major causes of soil pollution?

Soil pollution refers to the contamination of soil with substances that adversely affect its quality and health. This pollution can have detrimental effects on plant and animal life, as well as human health.

Causes of Soil pollution

Several factors contribute to soil pollution, with some major causes being:

1. Industrial Activities: Disposal of industrial waste, including heavy metals, solvents, and toxic chemicals, can lead to soil pollution. Accidental spills and leaks from industrial sites are common contributors.

2. Agricultural Practices: The use of pesticides, herbicides, and fertilizers in agriculture can introduce harmful chemicals into the soil. Overuse or improper disposal of these substances can result in soil degradation.

3. Improper Waste Disposal: Inadequate disposal of household waste, including plastic, electronic waste, and hazardous materials, can contaminate the soil. Open dumping and landfill sites without proper containment measures contribute significantly.

4. Mining Activities: Extraction of minerals and metals can release harmful substances into the soil, such as heavy metals and toxic chemicals, leading to soil pollution.

5. Deforestation: Clearing land for agriculture or urban development can disturb the natural balance of the soil, leading to erosion and degradation. The exposed soil is more susceptible to pollution.

6. Urbanization: Urban areas generate various pollutants, including heavy metals, oil, and chemicals from industries and transportation. Storm water runoff in urban areas can carry these pollutants into the soil.

7. Nuclear Accidents: Accidents involving the release of radioactive substances, such as nuclear power plant incidents, can result in soil contamination with long-lasting effects.

Q2. Briefly discuss persistent organic pollutants.

Persistent Organic Pollutants (POPs) are organic compounds that resist degradation through natural processes and can persist in the environment for extended periods. These pollutants typically bioaccumulate in living organisms, posing significant risks to ecosystems and human health.

Common examples include certain pesticides like DDT, industrial chemicals like polychlorinated biphenyls (PCBs), and unintentional by-products such as dioxins and furans. POPs can travel long distances through air and water, leading to their global distribution.

Due to their persistence, bioaccumulation, and potential for long-range transport, POPs have been a major focus of international environmental agreements. The Stockholm Convention, adopted in 2001, aims to eliminate or restrict the production and use of various POPs to protect human health and the environment. Efforts to manage and mitigate the impact of POPs involve regulation, monitoring, and sustainable alternatives to these substances.

Impacts of POPs

Persistent Organic Pollutants (POPs) can have detrimental effects on both the environment and human health. Here are some key impacts:

1. Environmental Impact

- **Bioaccumulation:** POPs tend to accumulate in the tissues of living organisms, with concentrations increasing as they move up the food chain.
- **Ecosystem Disruption:** POPs can disrupt ecosystems by affecting the reproductive, immune, and endocrine systems of various species.

2. Human Health Impact

- **Cancer Risk:** Certain POPs, like dioxins and PCBs, are known or suspected carcinogens, increasing the risk of cancer in exposed populations.
- **Endocrine Disruption:** Some POPs can interfere with the endocrine system, leading to hormonal imbalances and adverse reproductive effects.
- **Neurological Effects:** Exposure to certain POPs has been linked to neurodevelopmental problems, affecting cognitive function and behavior, especially in children.
- **Immune System Suppression:** POPs can weaken the immune system, making individuals more susceptible to infections and diseases.

3. Global Impact

- **Long-Range Transport:** POPs can travel over long distances through air and water, leading to global distribution and affecting regions far from their original sources.
- **Persistence:** The longevity of POPs means that their effects can persist over extended periods, even after measures are taken to reduce their production and use.

Q3. Discuss the principles of HPLC and its Application in environmental monitoring.

- High-Performance Liquid Chromatography (HPLC) is a powerful analytical technique used to separate, identify, and quantify components of a mixture based on their interactions with a stationary phase and a mobile phase. It is widely used in various fields, including chemistry, biochemistry, pharmaceuticals, and environmental monitoring.

Principle of HPLC:

The principle of HPLC is based on the differential affinities of the components in a sample for the stationary and mobile phases.

Stationary phase: The stationary phase is a solid or liquid material packed into a column. It has specific chemical properties and interacts differently with various components of the sample.

Mobile phase: The mobile phase is a solvent or a mixture of solvents that flows through the stationary phase, carrying the sample with it.

Sample injection: The sample is injected into the mobile phase, which is then forced through the column by a high-pressure pump.

Separation: As the mobile phase moves through the column, the different components of the sample interact with the stationary phase to varying degrees. This leads to the separation of the components based on their chemical properties and affinities for the stationary phase.

Detection: As each component emerges from the column, it passes through a detector that measures its concentration. This produces a chromatogram, which is a plot of detector response against time or volume.

Quantification: The area or height of each peak in the chromatogram corresponds to the quantity of the respective component in the sample. By comparing peak areas or heights to those of known standards, the concentration of each component can be determined.

Applications of HPLC in Environmental Monitoring:

Pesticide Residue Analysis: HPLC is used to detect and quantify pesticide residues in soil, water, and agricultural products. These residues can have detrimental effects on the environment and human health, so monitoring their levels is essential.

Water Quality Analysis: HPLC helps in identifying and measuring organic and inorganic pollutants in water bodies, including rivers, lakes, and groundwater. It allows environmental scientists to assess water quality and detect any potential hazards.

Air Pollution Monitoring: HPLC is employed to analyze air samples for volatile organic compounds (VOCs), polycyclic aromatic hydrocarbons (PAHs), and other air pollutants. This information aids in understanding air quality and sources of pollution.

Analysis of Heavy Metals: HPLC is used to determine the levels of heavy metals such as lead, mercury, and arsenic in environmental samples. These metals can be highly toxic and persistent in the environment, posing serious health risks.

Monitoring of Industrial Effluents: HPLC is utilized to assess the presence of organic compounds and harmful chemicals in industrial effluents before they are discharged into the environment. This helps in enforcing environmental regulations.

Analysis of Environmental Contaminants: HPLC is valuable in identifying and quantifying various environmental contaminants, including phenols, herbicides, pharmaceutical residues, and industrial chemicals.

Q4. Briefly describe the composition of a typical soil.

A typical soil is composed of mineral particles, organic matter, water, and air. The relative proportions of these components vary, and soil is often classified into different horizons or layers based on these variations. Here is a general breakdown of the composition of a typical soil:

1. Mineral Particles

Sand (largest particles): Typically ranging from 0.05 to 2.0 millimeters in diameter.

Silt (medium-sized particles): Smaller than sand, ranging from 0.002 to 0.05 millimeters.

Clay (smallest particles): Particles are less than 0.002 millimeters in diameter.

2. Organic Matter

Decomposed plant and animal material: This organic matter contributes to soil fertility and structure. It includes humus, which is rich in nutrients and enhances the soil's ability to retain water.

3. Water

Pore spaces between soil particles: These spaces are filled with water to varying degrees, depending on factors such as precipitation, drainage, and soil texture.

4. Air

Pore spaces between soil particles: In addition to water, soil contains air in the spaces between particles. This is essential for the respiration of plant roots and soil organisms.

5. Microorganisms

Bacteria, fungi, and other microbes: These contribute to the decomposition of organic matter, nutrient cycling, and other biological processes crucial for soil health.

6. Inorganic Material

Weathered rocks and minerals: The mineral composition of the soil is influenced by the parent material and geological processes. It provides essential nutrients for plant growth.

7. Nutrients

Essential elements for plant growth: Nutrients such as nitrogen, phosphorus, potassium, calcium, and others are present in varying amounts, influencing the soil's fertility.

8. pH

Acidity or alkalinity level: The pH of the soil affects nutrient availability and microbial activity. Soils can be acidic, neutral, or alkaline.

Q5. Write a note on legislation aspects of environmental pollution.

Legislation plays a crucial role in addressing and mitigating environmental pollution. Governments worldwide have enacted laws and regulations to safeguard the environment and human health. Key aspects of environmental pollution legislation include:

Emission Standards: Legislation often establishes permissible limits for pollutant emissions from industrial facilities, vehicles, and other sources. Emission standards aim to control and reduce the release of harmful substances into the air, water, and soil.

Waste Management: Laws govern the proper disposal, treatment, and handling of various types of waste. This includes hazardous waste, solid waste, and electronic waste. Compliance with waste management regulations helps prevent soil and water contamination.

Water Quality Protection: Legislation addresses the protection and conservation of water resources. It sets standards for water quality, regulates discharges into water bodies, and promotes sustainable water management practices.

Land Use Planning: Regulations often guide land use to prevent environmental degradation. Zoning laws may restrict certain activities in environmentally sensitive areas, preserving ecosystems and preventing pollution.

Biodiversity Conservation: Legislation may include provisions for the protection of endangered species and ecosystems. Preserving biodiversity contributes to ecological balance and resilience against environmental threats.

Liability and Enforcement: Legal frameworks establish liability for environmental damage, holding polluters accountable for the harm caused. Enforcement mechanisms, such as fines and penalties, incentivize compliance with environmental laws.

Adaptation and Resilience Planning: In response to climate change, legislation may include provisions for adaptation and resilience planning. This involves

strategies to cope with changing environmental conditions and reduce vulnerabilities.

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2022

SHORT QUESTIONS

Q1. What is HUMUS?

Humus is a dark, organic material that forms in soil as a result of the decomposition of plant and animal matter. It is rich in nutrients and serves as a vital component of healthy soil, enhancing its fertility and structure. Humus improves water retention in soil and promotes aeration, making it ideal for plant growth. It also helps to sequester carbon from the atmosphere, playing a role in mitigating climate change.

Q2. How soil air is important for fertility?

Soil air is essential for soil fertility because it provides oxygen to plant roots, enabling them to carry out respiration and absorb nutrients more efficiently. Adequate oxygen levels in the soil prevent root suffocation and promote healthy root growth. Soil air also helps in the decomposition of organic matter by providing the necessary oxygen for soil microorganisms, which release nutrients essential for plant growth.

Q3. Suggest some control measures for water erosion.

- Planting grass, trees, or other vegetation helps to stabilize the soil and reduce erosion.
- Constructing terraces on sloped land can reduce the speed of runoff and trap sediment.
- Installing silt fences or sediment basins can help to capture eroded soil particles before they reach water bodies.
- Develop and maintain proper drainage systems to redirect excess water away from vulnerable areas and prevent erosion.

Q4. What type of pollutants can be analyzed by AAS?

Atomic Absorption Spectroscopy (AAS) is a technique primarily used for the analysis of metal elements in various samples. It is particularly effective for analyzing trace and ultra-trace levels of metals. Some of the pollutants that can be analyzed by AAS include:

Heavy Metals: AAS is commonly employed to analyze heavy metals such as lead (Pb), cadmium (Cd), mercury (Hg), arsenic (As), chromium (Cr), and others in environmental samples like water, soil, and air.

Metalloids: Elements like arsenic, which has metalloid properties, can be analyzed by AAS. Arsenic is often of concern due to its toxicity, and AAS provides a sensitive method for its quantification.

Nutrient Metals: AAS is used in the analysis of essential nutrient metals like iron (Fe), zinc (Zn), copper (Cu), and manganese (Mn) in biological samples, soils, and food products.

Pharmaceuticals: AAS can be applied to analyze metal impurities in pharmaceuticals. It helps ensure that pharmaceutical products meet regulatory standards regarding metal content.

Water Quality Monitoring: AAS is widely employed in water quality monitoring to analyze metals in drinking water, wastewater, and natural water bodies.

Q5. What is meant by AFLATOXINS?

Aflatoxins are highly toxic and carcinogenic secondary metabolites produced by certain moulds, primarily *Aspergillus* species, that contaminate crops like peanuts, corn, and tree nuts. These mycotoxins pose serious health risks when ingested or inhaled. They can contaminate food products at various stages, from cultivation to storage, and are especially prevalent in warm and humid climates.

Q6. How availability of phosphorus is affected by soil pH?

The availability of phosphorus in soil is significantly affected by soil pH. In acidic soils with a pH below 6, phosphorus tends to become less available because it forms insoluble compounds with iron and aluminium. Conversely, in alkaline soils with a pH above 7.5, phosphorus can also become less available due to precipitation with calcium and magnesium. The optimal pH range for phosphorus availability is typically between 6 and 7. In this range, phosphorus is most soluble and accessible to plants.

Q7. What is chemical speciation?

Chemical speciation in environmental chemistry refers to the process of identifying and understanding the different chemical forms or species that a particular element or compound can exist in within a given environmental system. Speciation is crucial because different chemical species of the same element can have varying toxicity, mobility, and reactivity.

For example, in the case of heavy metals like mercury, speciation is essential because the toxic effects of methylmercury, a specific chemical form, can be far

more harmful than other mercury compounds. Speciation studies help in assessing environmental risks, understanding biogeochemical cycles, and developing effective remediation strategies for pollutants

Q8. The propose the reclamation of basic soil.

Conduct a soil test to determine its pH and nutrient levels. Apply acidic amendments such as sulfur or peat moss to lower the pH. Monitor the soil pH regularly and adjust as needed to maintain a slightly acidic range (around 6.0-7.0). Introduce organic matter like compost to improve soil structure and nutrient content. Implement crop rotation and proper irrigation practices to sustainably manage the soil's health over time.

Q9. What is principle of UV spectroscopy?

Q10. Mention the salient features of HPLC.

High-Performance Liquid Chromatography (HPLC) is a widely used analytical technique with several salient features:

High Resolution: HPLC provides high resolution due to the use of small particle size in the stationary phase, resulting in efficient separation of analytes.

Sensitivity: It is a highly sensitive technique, allowing the detection and quantification of trace amounts of compounds in a sample.

Precision and Accuracy: HPLC provides precise and accurate results, making it suitable for both qualitative and quantitative analyses.

Automation: HPLC systems are often automated, allowing for increased sample throughput, reduced human error, and enhanced reproducibility.

Quantitative Analysis: It is widely used for quantitative analysis due to its ability to provide accurate and reproducible concentration measurements.

Q11. What are the primary and secondary minerals of soil?

Primary minerals in soil are those originally formed during the rock's geological formation and include minerals like quartz, feldspar, and mica. They have relatively stable compositions.

Secondary minerals in soil form through various weathering processes over time and include minerals like clay minerals (e.g., kaolinite, montmorillonite), iron oxides, and aluminium silicates. They often have more variable compositions and can influence soil properties.

Q12. What are the limitations of GC?

Gas chromatography (GC) has some limitations, including:

Volatility of Samples: GC is most effective for volatile compounds. If a sample contains non-volatile or semi-volatile components, they may not be well-suited for analysis using GC.

Thermal Stability: Some compounds may decompose at the high temperatures used in GC, leading to inaccurate results or loss of analytes.

Sample Size: GC typically requires a small sample size. If the sample size is too large, it might overload the column or affect the separation efficiency.

High Initial Cost: The initial cost of a GC system, including the instrument and accessories, can be relatively high.

Q13. What is DIRTY DOZEN?

The "Dirty Dozen" is a term often used to refer to a list of twelve fruits and vegetables that are commonly found to have higher pesticide residues when conventionally grown. These include strawberries, spinach, kale, nectarines, apples, grapes, peaches, cherries, pears, tomatoes, celery, and potatoes.

Consumers are advised to consider buying these items organic, if possible, to reduce pesticide exposure. The list is annually updated by the Environmental Working Group (EWG) based on pesticide residue data from the USDA.

Q14. Where is the word SOIL from?

The word soil is derived from the Latin word 'solum' meaning earthly material in which plants grow. Soil is made up of four components: Inorganic materials such as minerals that are derived from parent rocks. Organic materials such as humus are derived from dead organisms by their decomposition.

Q15. What is the importance of clay in the nutrient-holding capacity of soil?

Clay plays a crucial role in the nutrient-holding capacity of soil due to its small particle size and unique properties:

High Surface Area: Clay particles have a large surface area per unit volume, providing more sites for nutrient adsorption.

Cation Exchange Capacity (CEC): Clay has a high CEC, meaning it can hold and exchange cations (positively charged ions like calcium, magnesium, and potassium), allowing plants to access essential nutrients.

Nutrient Retention: Clay retains nutrients through electrostatic forces, preventing them from leaching away with water.

Slow Release: Nutrients held by clay are released slowly over time, providing a consistent supply to plants.

Improved Fertility: Clay-rich soils tend to have better nutrient availability, enhancing their suitability for agriculture and plant growth

LONG QUESTIONS

Q1. Describe liquid and gaseous components of soil.

The liquid and gaseous components of soil play essential roles in supporting plant growth and maintaining soil health.

Soil Water: Water is in liquid or gaseous form and may occupy pore spaces not occupied by gases.

- 1) Water acts as a medium where nutrients from the minerals and decaying organic matter are released and made available to plants. Root water is also an effective temperature regulator of soil giving the living organisms a more stable environment to grow and reproduce.

Soil Solution: The water dissolves salts and makes soil solution. The soil solution is important as a medium for supplying different essential nutrients between the soil solids and the soil solution and then between the soil solution and the plants.

Soil Moisture: Water contained in the soil is called soil moisture. The two basic forces for water retention in soil are cohesive and adhesive forces.

Cohesive Forces: The forces of attraction present between the molecules of water are called cohesive forces.

Adhesive Forces: The forces of attraction between the water molecule and solid surface are called adhesive forces.

Function of Water

These are four important functions of water in plant growth.

- It is the major constituent of plant protoplasm (85-95% water)
- It is essential for photosynthesis and conversion of starches to sugars.
- Soil solution is important as a medium for the supply of nutrients to growing plants
- It provides plant turgidity, which maintains the proper form and position of plant parts to capture sunlight.
- The soil water is a great regulator of physical, chemical and biological activities in the soil plants absorb some water through leaf stomata but most of the water used by plants is absorbed by the roots from the soil.

The soil water may be

- i) Gravitational water**
- ii) Capillary water**
- iii) Hygroscopic water**

SOIL AIR:

- Air spaces are found in between the soil particles, and hold air which constitutes a gaseous system of the soil. Gases occupy these spaces. The most commonly found gases in soil include N₂, O₂, and CO₂. The earth's atmosphere is composed of mainly 78% N₂, 21% O₂ and 1% CO₂. The respiration of plant roots and microbes generally increases the level of O₂ and CO₂ in the atmosphere. Soil gases are essential to the life cycles of soil animal organisms and plants.

Soil Aeration: The exchange of CO₂ and O₂ gases between the soil pore space and the aerial atmosphere is called soil aeration. A well-aerated soil is one that contains a sufficient amount of gases available to growing aerobic organisms.

Poor Soil Aeration: Poor soil aeration results in the accumulation of CO₂, which causes the formation of H₂S, formic acid acetic acid oxalic acid and other toxic acids. Because of these toxic acids soil organisms (both micro and macro-organisms) are killed. Excess of CO₂ is also responsible for the decrease in PH of the soil which prevents nutrients from reaching into the plants.

Characteristics of Soil Air

- *Due to poor creation*
- The development of plants' roots is restricted
- Absorption of water and nutrients decreased
- Toxic substances are formed
- Produce different types of diseases
- The rate of organic matter decomposition decreases due to decreased microbial activity under reduced O₂ contents.

Q2. How mining is contributing to soil pollution?

Mining can contribute to soil pollution through various mechanisms:

Chemical Contamination: Mining operations often involve the extraction of minerals and ores, which may contain heavy metals and other toxic substances. These substances can be released into the soil, contaminating it and making it unsuitable for agriculture or other uses.

Acid Mine Drainage: In many cases, mining exposes sulfide minerals to air and water, leading to a chemical reaction that produces acid mine drainage (AMD).

AMD can lower the pH of soil, making it inhospitable for plant growth and causing the release of heavy metals from the soil.

Soil Erosion: Mining activities, especially open-pit mining, can result in extensive soil erosion. This not only displaces fertile topsoil but can also carry pollutants into nearby water bodies, further impacting the environment.

Deforestation: Clearing land for mining operations often involves cutting down forests and vegetation. This can lead to soil erosion, loss of biodiversity, and disruption of ecosystems.

Mining Waste Disposal: The disposal of mining waste, such as tailings, can contaminate soil. These waste materials may contain heavy metals, chemicals, and other pollutants that can leach into the surrounding soil.

Land Subsidence: Underground mining can result in land subsidence, causing the soil to collapse and disrupt natural drainage patterns. This can lead to the accumulation of pollutants in the subsided areas.

Airborne Contaminants: Dust and emissions from mining activities can settle on nearby soil, introducing contaminants into the soil.

Mining practices and regulations vary widely, so the extent of soil pollution caused by mining depends on factors like the type of minerals being extracted, the location of the mine, and the environmental regulations in place. To mitigate these issues, responsible mining practices and effective environmental management are crucial.

Q3. Write down the salient features of POPs.

- POPs, or Persistent Organic Pollutants, are a group of chemical compounds known for their long-lasting environmental and health effects.

Here are some of their salient features:

Persistence: POPs are highly resistant to degradation processes, making them persistent in the environment. Some can last for decades or even longer.

Bioaccumulation: These chemicals tend to accumulate in living organisms, especially in the fatty tissues of animals, as they move up the food chain.

Long-Range Transport: POPs can be transported over long distances through air and water, impacting regions far from their original sources.

Toxicity: Many POPs are toxic to humans and wildlife, leading to a wide range of health problems, including cancer, reproductive disorders, and developmental issues.

Widespread Distribution: POPs have been found in remote areas, including the Arctic and Antarctic, highlighting their global reach.

Regulation: International agreements, such as the Stockholm Convention, aim to reduce and eliminate the production and use of POPs to protect human health and the environment.

Diverse Chemicals: POPs encompass a broad range of organic compounds, including pesticides (e.g., DDT), industrial chemicals (e.g., PCBs), and unintentional by-products (e.g., dioxins).

Historical Use: Many POPs were widely used in the past for various purposes, but their negative impacts on the environment and health led to restrictions and bans.

Monitoring and Management: There are ongoing efforts to monitor and manage POPs to mitigate their harmful effects on ecosystems and human populations.

These features highlight the significant environmental and health concerns associated with POPs and the need for continued efforts to reduce their presence and impact entering the human food chain through contaminated seafood.

Q4. Describe any 6 principles of GREEN chemistry.

Q5. Describe the importance of SPECTROSCOPIC techniques in environmental monitoring.

- Spectroscopy plays a vital role in environmental monitoring due to its ability to identify and quantify various substances present in the environment. It involves the interaction of electromagnetic radiation with matter, providing valuable information about the composition and properties of environmental samples.

Here are some key reasons why spectroscopic techniques are essential in environmental monitoring:

Identification of Pollutants: Spectroscopic techniques such as UV-visible spectroscopy, Infrared (IR) spectroscopy, and Nuclear Magnetic Resonance (NMR) spectroscopy allow scientists to identify different pollutants present in environmental samples. By analyzing the unique spectra produced by various substances, researchers can determine the presence of harmful chemicals, heavy metals, and organic pollutants.

Quantitative Analysis: Spectroscopic methods enable quantitative analysis, allowing researchers to measure the concentration of pollutants accurately. This information is critical for assessing pollution levels and determining whether they comply with environmental regulations and standards.

Non-Destructive Analysis: Spectroscopic techniques are often non-destructive, meaning they do not alter the sample during analysis. This advantage allows scientists to perform multiple measurements on the same sample, providing more comprehensive data without the need for excessive sampling.

Real-Time Monitoring: Some spectroscopic techniques, such as UV-visible spectroscopy and fluorescence spectroscopy, can be applied in real-time and even in situ. This capability enables continuous monitoring of environmental parameters, providing instant feedback on pollution levels and facilitating rapid responses to emerging environmental issues.

Detection of Trace Amounts: Many pollutants and contaminants in the environment exist in trace amounts, which can be challenging to detect using conventional methods. Spectroscopic techniques are highly sensitive and can detect even low concentrations of substances, making them invaluable for monitoring subtle changes in environmental conditions.

Monitoring Chemical Reactions: Spectroscopic techniques help monitor chemical reactions occurring in the environment, such as the transformation and degradation of pollutants. Understanding these processes is vital in developing effective remediation strategies.

Faster Analysis: Spectroscopic methods often provide rapid results, allowing for quicker decision-making and timely responses to environmental issues. This speed is crucial in emergencies and critical situations.

Research and Policy Development: The data obtained through spectroscopic techniques contribute to scientific research and help in the development of environmental policies and regulations. These policies are essential in protecting the environment and human health.



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